Morphic Studio

Document 31 — Monitoring & Analytics (Expanded Specification)

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Version: 2.0

Status: Expanded Specification

Priority: Core Operations Infrastructure

Purpose

The Monitoring & Analytics System provides continuous visibility into the health, performance, reliability, and usage of Morphic Studio.

It enables developers and administrators to detect issues early, optimize system performance, understand user behavior, improve AI workflows, and make informed product decisions.

Monitoring is not only about detecting failures—it is about understanding how the platform evolves over time.

Vision

The platform should answer questions like:

- Is the platform healthy?
- Are AI workflows completing successfully?
- Are users experiencing delays?
- Which features are most valuable?
- Which AI provider performs best?
- Where are failures occurring?
- How can the platform improve?

Core Philosophy

You cannot improve what you cannot observe.

Every major system should produce measurable information that helps improve reliability, usability, and creative productivity.

Monitoring Objectives

The platform shall:

- Monitor application health.
- Monitor AI workflows.
- Monitor infrastructure.
- Monitor storage.
- Monitor exports.
- Monitor user experience.
- Record operational metrics.
- Support troubleshooting.

Monitoring Domains

The system monitors:

Application

- API health
- Backend services
- Frontend performance
- Database connectivity

AI Systems

- Story analysis
- Character generation
- Comic generation
- Animation generation
- Voice generation
- Audio generation
- Export workflows

Infrastructure

- CPU usage
- Memory usage
- Storage capacity
- Network availability
- Response times

User Experience

- Page load time
- Workflow completion time
- Upload speed
- Download speed
- Rendering time
- Export duration

Business Metrics

- Active users
- Projects created
- Assets generated
- Stories created
- Comics exported
- Animations exported
- Storage usage

System Health Dashboard

The Operations Dashboard should display:

System Status

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API Status

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Database Status

↓

Storage Status

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AI Provider Status

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Export Queue

↓

Background Jobs

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Recent Errors

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Notifications

Administrators should immediately understand platform health.

AI Workflow Monitoring

Every AI workflow records:

- Workflow ID
- Project
- User
- AI Provider
- Model Used
- Start Time
- End Time
- Duration
- Status
- Tokens Used (if applicable)
- Errors
- Output Location

This supports debugging, optimization, and future cost analysis.

Performance Metrics

The platform should measure:

- Request latency
- Database query time
- Asset retrieval time
- Upload duration
- Rendering duration
- Export duration
- Workflow completion time

Historical trends should be retained for analysis.

Usage Analytics

Anonymous or privacy-respecting analytics may include:

- Most used features
- Most common workflows
- Average project size
- AI workflow frequency
- Export formats
- Device types
- Session duration

These insights help prioritize future development.

Alerting

Alerts should be generated for:

- Service outages
- High error rates
- Failed backups
- AI provider failures
- Storage capacity thresholds
- Repeated authentication failures
- Export failures

Alerts should include sufficient information for diagnosis.

Logging Integration

Monitoring integrates with:

- Application logs
- Workflow logs
- Audit logs
- Security events
- Deployment logs
- Performance logs

Together they provide a complete operational picture.

User Stories

As a developer, I want to know when something breaks before users report it.

As an administrator, I want visibility into platform health.

As a creator, I want confidence that my long-running AI workflows are progressing normally.

Functional Requirements

The Monitoring System shall:

- Collect metrics.
- Display dashboards.
- Generate alerts.
- Record workflow history.
- Track AI performance.
- Support troubleshooting.
- Preserve operational history.

Dashboard Components

The Monitoring Dashboard includes:

- System Overview
- Service Status
- AI Activity

- Workflow Queue
- Error Timeline
- Storage Metrics
- Performance Graphs
- Active Sessions
- Notifications
- Recent Deployments

Database Entities

Core entities include:

- Metric
- HealthCheck
- WorkflowMetric
- PerformanceRecord
- Alert
- Notification
- ServiceStatus
- DashboardSnapshot

API Considerations

Potential endpoints include:

- Health Status
- Metrics
- Performance History
- Workflow Statistics
- Alert History
- Dashboard Summary
- Service Availability

Non-Functional Requirements

The Monitoring System should:

- Have minimal performance impact.
- Scale with platform growth.
- Support historical reporting.
- Remain reliable during failures.

- Preserve operational data securely.

Error Handling

If monitoring services become unavailable:

- Continue core platform operations.
- Buffer metrics where possible.
- Resume reporting automatically.
- Notify administrators.

Monitoring failures should never prevent creators from working.

Future Expansion

Future versions may include:

- AI-assisted anomaly detection.
- Predictive maintenance.
- Cost optimization dashboards.
- User satisfaction analytics.
- AI model performance comparisons.
- Capacity forecasting.
- Personalized productivity insights.
- Real-time collaboration analytics.

Acceptance Criteria

The Monitoring & Analytics System succeeds when developers, administrators, and future teams can clearly understand platform health, user behavior, AI performance, and operational trends without disrupting creators' workflows.

Key Principle

Monitoring transforms data into understanding. By continuously observing the platform, Morphic Studio can become more reliable, efficient, and creator-focused with every iteration.

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